Home Rehabilitation and health Monitoring system

—WellSpan

Project Presentation

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Contents

Project Background and Objectives

- Introduce the project’s core vision
- Clarify the R&D direction and expected outcomes.

Project Management and Division of Labor

Introduce the team collaboration model and members' responsibilities to ensure the project progresses efficiently

Hardware System Design

- Present the circuit schematic and PCB layout
- Explain the rationale behind the core hardware selection.

Software architecture and algorithms

Analyze the software architecture and discuss the implementation details of key algorithms.

Upper computer and structure

Introduce the functions of the human-computer interaction interface and briefly describe the design plan of the equipment's mechanical structure.

Summary and Outlook

Review the project's achievements and shortcomings, and plan for future iterations and commercial value.

Project Background and Problem Statement

Medical resource gap

There are not enough community medical workers, so they cannot do high-frequency and regular health monitoring on patients.

Equipment has poor adaptability

Medical equipment is expensive, consumer equipment has simple functions, and lacks professional monitoring functions for rehabilitation scenes.

Rehabilitation compliance is low

Medical workers cannot remotely and exactly assess patients' rehabilitation conditions, so rehabilitation plans are hard to change dynamically.



居家康复场景

Core goal of the project

Sital Signs Monitoring
Real-time tracking of heart rate and blood oxygen

Rehabilitation environment monitoring
Real-time monitoring of temperature and humidity

Limb balance monitoring
Judge balance ability through level meter

康复训练互动
转移注意力，降低病人的心理压力

Additional engineering goals

Bluetooth wireless data transmission and visual display

Finish the whole development from bottom hardware to top software

Project Management and Team Division of Work

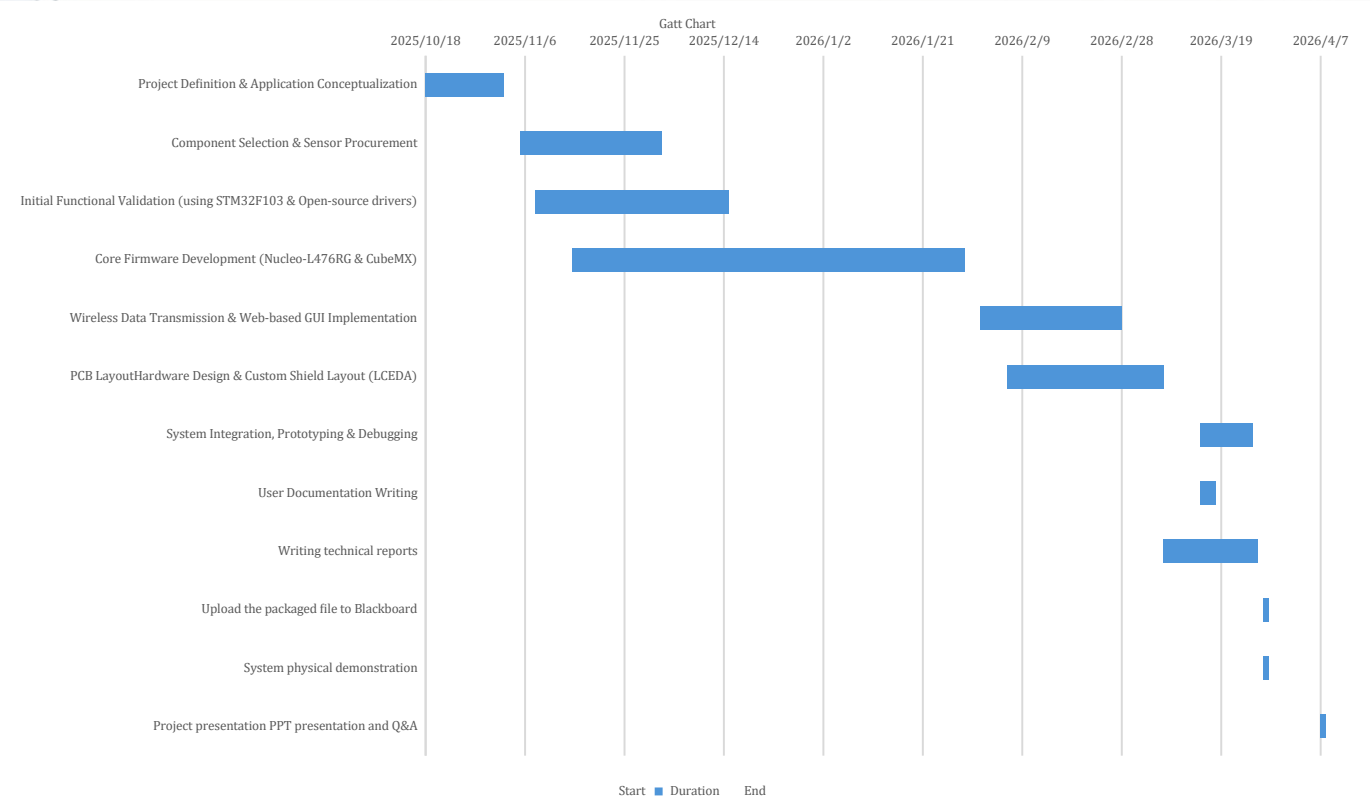


Figure 1: Gantt chart of the whole project progress

Management framework ：

Strictly follow the Prince2 small project management framework to make sure the project is delivered on time and with good quality.

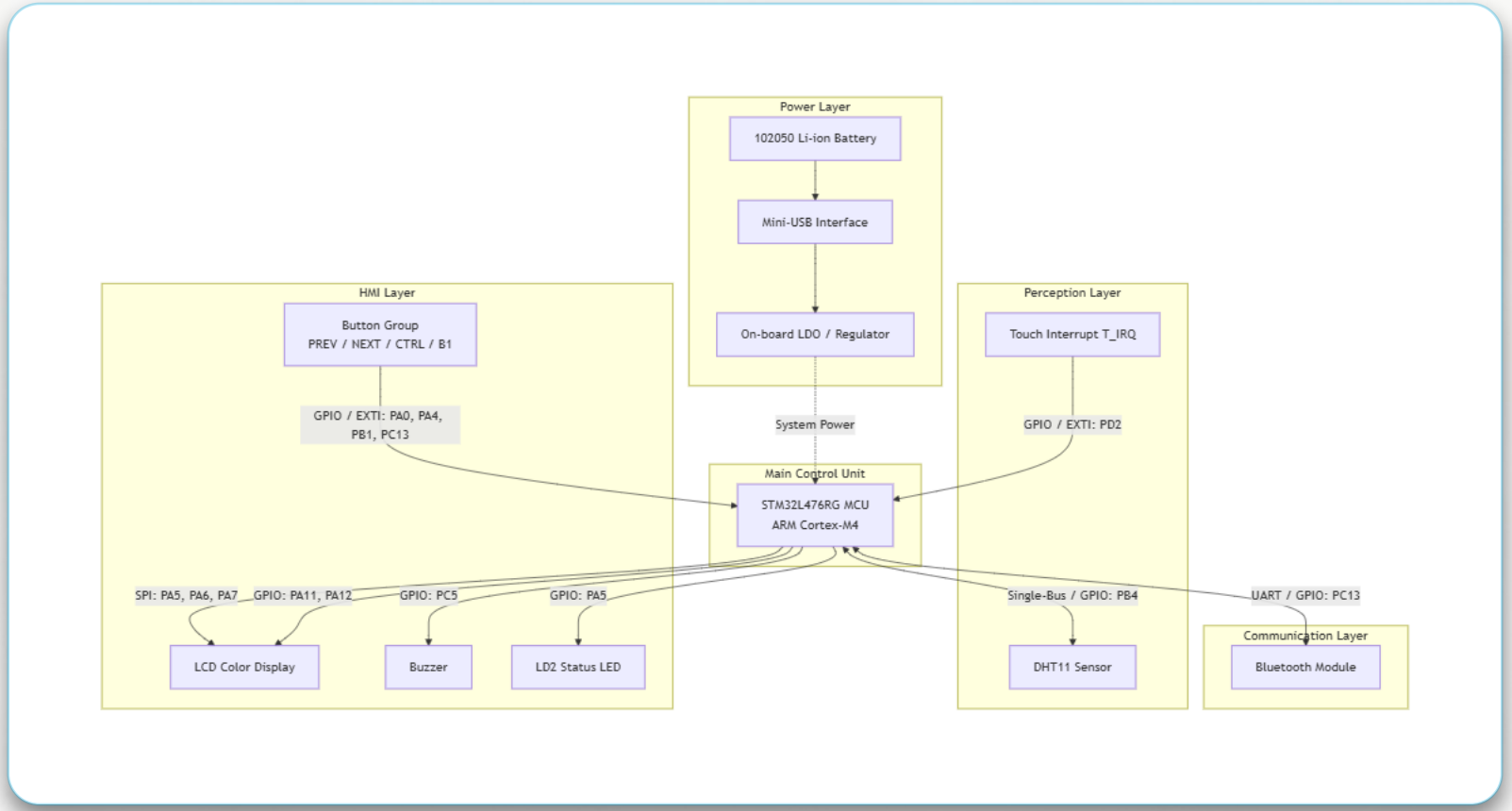
Lowlife555 (Person in Charge of Software and Algorithms)

Be responsible for embedded software development, core algorithm implementation and PC upper computer software

Mo Yan (Person in charge of hardware)

Be responsible for schematic and PCB design, hardware test and project report writing

Overall architecture of the hardware system



Core design principles

Only use STM32L476RG as the main control unit. Adopt layered design with clear responsibilities and boundaries to ensure system stability and scalability

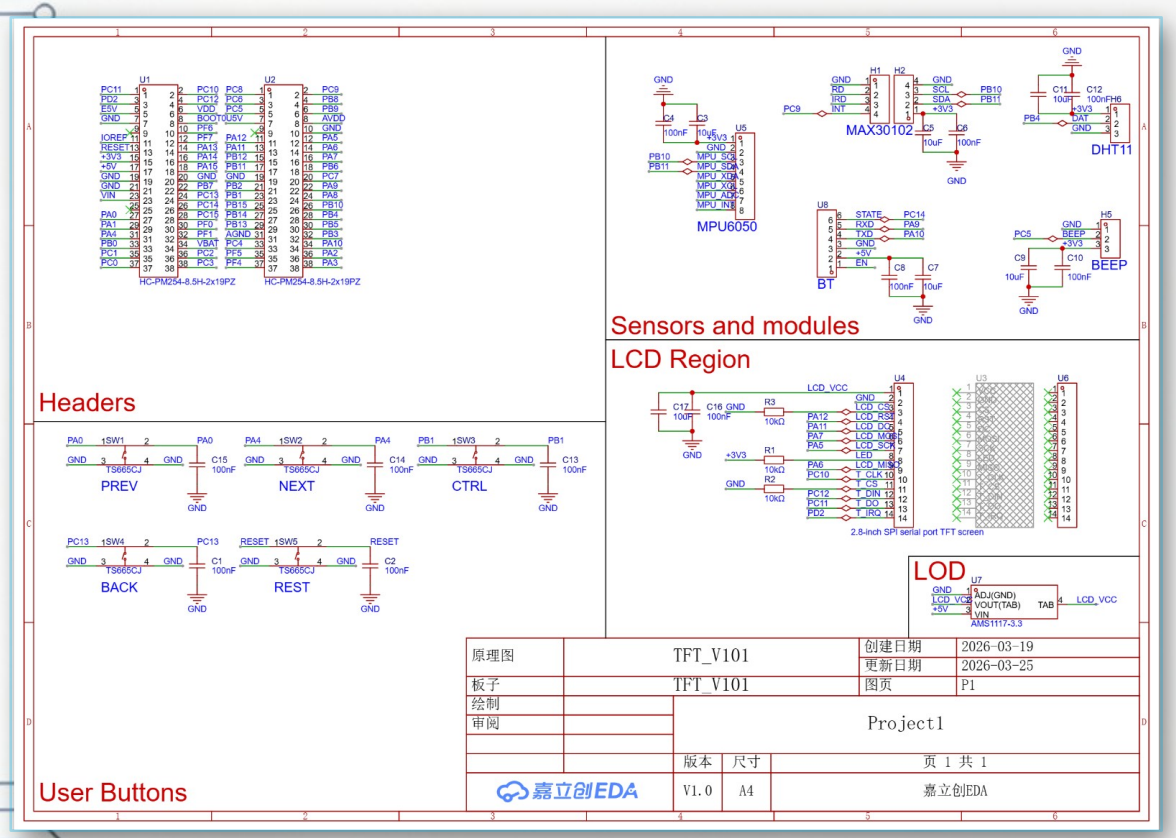
The whole hardware link

Main control core → sensor acquisition module → human-computer interaction module → Bluetooth wireless communication module → power module, to realize the closed loop of the whole link

Core component selection and compliance

Device Model	Core Functions	Core reasons for selection	Course compliance
STM32L476RG	Main controller	The course requires it, its performance meets the needs, and it has many external devices.	conform
MAX30102	Heart rate and blood oxygen monitoring	It is made for wearable devices, with low power use and high sensitivity.	conform
MPU6050	Motion / posture monitoring	6-axis IMU with built-in DMP, full functions	conform
DHT11	Temperature and humidity monitoring	Commonly used in labs, easy to connect	conform
ILI9341	Display	Big color screen, fits user needs	conform
HC-05	Bluetooth communication	Stable, works with PC, fits monitoring needs	conform

Key points of schematic design



Module design

All circuits are tested on breadboard prototypes to make sure the logic is correct.

Power management

LCD power uses AMS1117-3.3 voltage regulator chip to keep power stable.

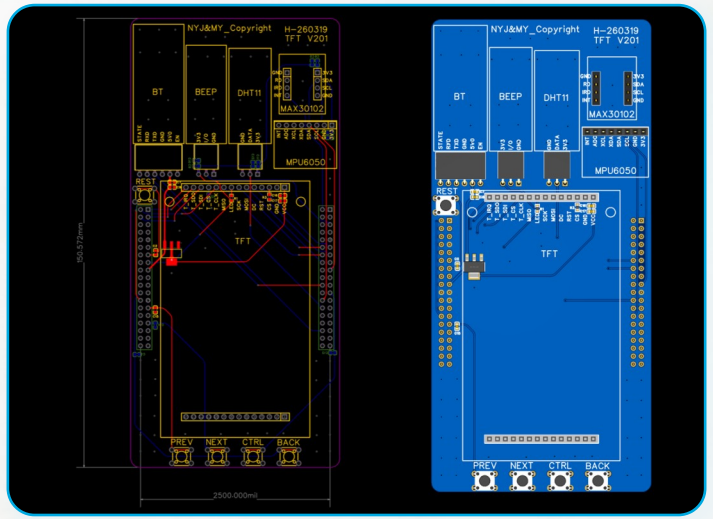
I2C bus multiplexing

MAX30102 and MPU6050 share I2C bus, make PCB wiring less complex

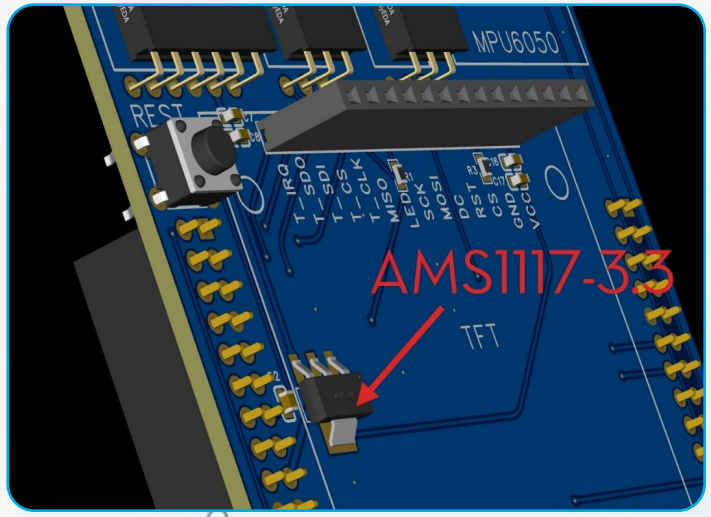
Active alarm circuit

Design buzzer drive circuit, with fall detection algorithm to realize real-time alarm

PCB design and engineering implementation



PCB expansion board design layout



ILI9341 区域 3D 预览

Design Principles

• Modular partitioning minimizes signal paths. Thermal isolation and electromagnetic shielding ensure stability.

• Layout adapted to ergonomic operating habits

Key optimization

• Using curved traces improves high-frequency signal integrity.

• Sensors are placed close together to shorten bus length.

• Centered screen design optimizes visual interaction experience

Evolutionary process

Through three rapid iterations, from prototype verification to final product optimization, every detail was ensured to meet engineering standards.

Software overall architecture

Hardware Abstraction Layer

Low-level driver encapsulation based on HAL library

Algorithm and Data Processing Layer

Attitude calculation, heart rate and blood oxygen core calculation

Peripheral driver layer

Achieve independent driving for sensors, screens, etc.

Application and Task Scheduling Layer

UI Interaction Logic and System Task Management

Scheduling: Time-Slice Round-Robin

Use a main loop time-slice polling strategy to allocate runtime cycles based on the real-time nature of functions. Ensure the responsiveness of high-priority tasks such as fall detection and game interaction.

Core algorithm implementation: MPU6050

Core Function

It enables attitude angle calculation, accurate step counting, fall detection, and provides data for gravity-sensing games.

Algorithm Implementation Logic

Step counting process: Combined acceleration calculation → Low-pass filtering → Peak detection → Time repetition prevention.

Fall detection: An alarm is triggered when the acceleration on any axis is greater than 3G or when both the X and Y axes are greater than 2G.

Actual performance indicators

The false alarm rate for fall detection is controlled **< 5%**， Response speed **< 500ms** 。

Motion posture monitoring

Based on an accelerometer fusion algorithm, it reconstructs human motion and supports step counting.

Safety early warning system

Precise impact monitoring triggers an audible and visual alarm immediately upon detecting fall characteristics, ensuring user safety.

Motion-sensing interactive games

By using posture angle data to drive gravity-sensing games such as maze balls, rehabilitation training and entertainment can be organically combined.

Core algorithm implementation: MAX30102

Signal processing workflow

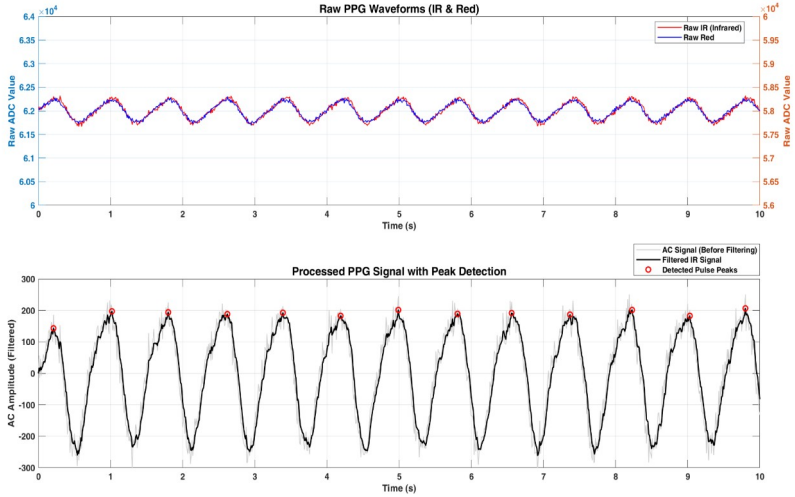
Communication debugging: Shares the I2C bus with MPU6050 to read raw PPG data.

Signal preprocessing: IIR filtering removes baseline drift, and low-pass filtering reduces noise.

Heart rate calculation: Peak detection + moving average algorithm to determine instantaneous heart rate

Blood oxygen calculation: Based on the Lambert-Beer law, the peak-to-peak absorption ratio is calculated.

Algorithm Performance & Accuracy



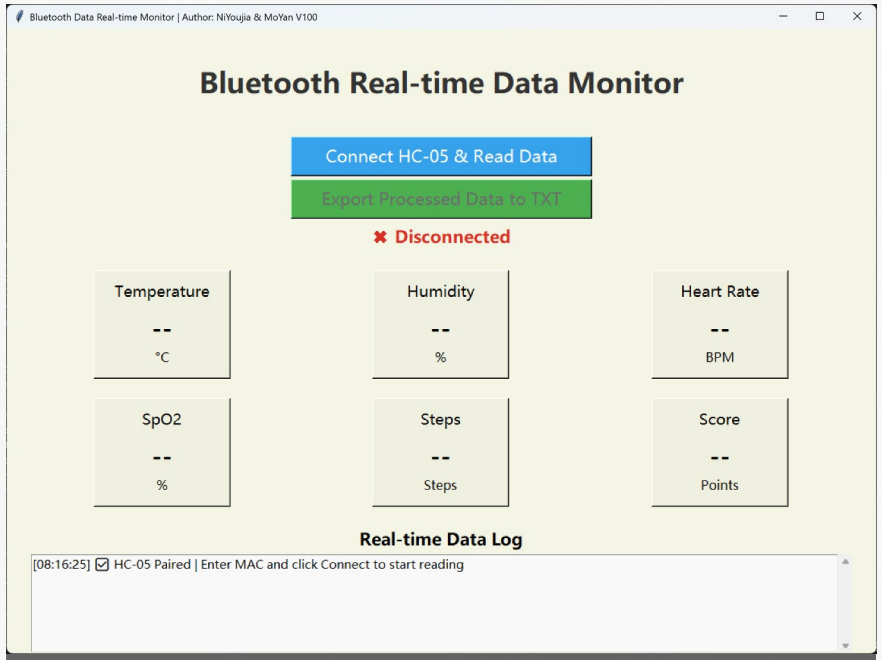
PPG signal waveform processing:
The algorithm eliminates motion artifacts and high-frequency noise, and extracts clean signal features.

Heart Rate error
±5 bpm

Blood oxygenation error
±2 %

Data source: Average error of 20 samples measured in a laboratory environment, with a small difference compared to consumer-grade heart rate and blood oxygenation monitoring equipment. (Samsung Galaxy Watch)

PC host computer and data logging function



A visual monitoring terminal developed using Python.

Bluetooth communication
基于 RFCOMM 协议与 HC-05 模块建立高可靠连接，确保数据不丢包。

Real-time visualization of multiple data
动态刷新并展示心率、血氧、温湿度及步数等关键指标，直观掌握状态。

Real-time log display
实时打印系统运行日志与通信状态，便于开发调试与异常快速排查。

Data Recording and Management
自动保存历史数据至本地，支持一键导出为 .txt 格式文件，便于归档分析。

User Guide and Feature Demonstration

Operation logic: Supports both physical buttons and touchscreen operation to suit different user habits.

Temperature and humidity

After entering the monitoring interface, the system automatically refreshes the current environmental data.

Heart rate and blood oxygen

Place your finger lightly on the top sensor and remain still for 15 seconds to wait for the result to be generated.

Balance ability assessment

By monitoring bubble displacement through a built-in level, the body's balance can be scientifically assessed.

Interactive games

Combining physical buttons with gravity sensing technology makes rehabilitation training more fun.

Fall detection

The algorithm runs automatically in the background and immediately triggers an audible and visual alarm when a fall is detected.

Wireless transmission

The Bluetooth function can send data to the host computer in real time.

Compliance Statement and Risk Warning

Compliance Statement

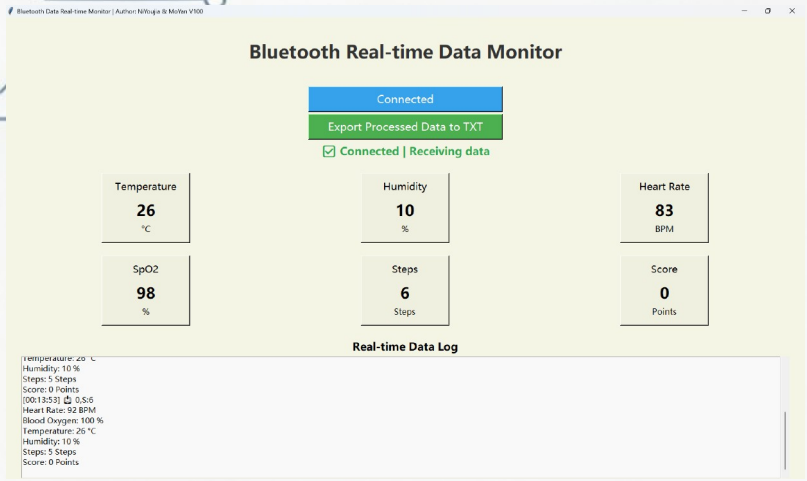
This device is for rehabilitation aid reference only. **Not qualified for medical diagnosis**, cannot replace professional medical equipment.

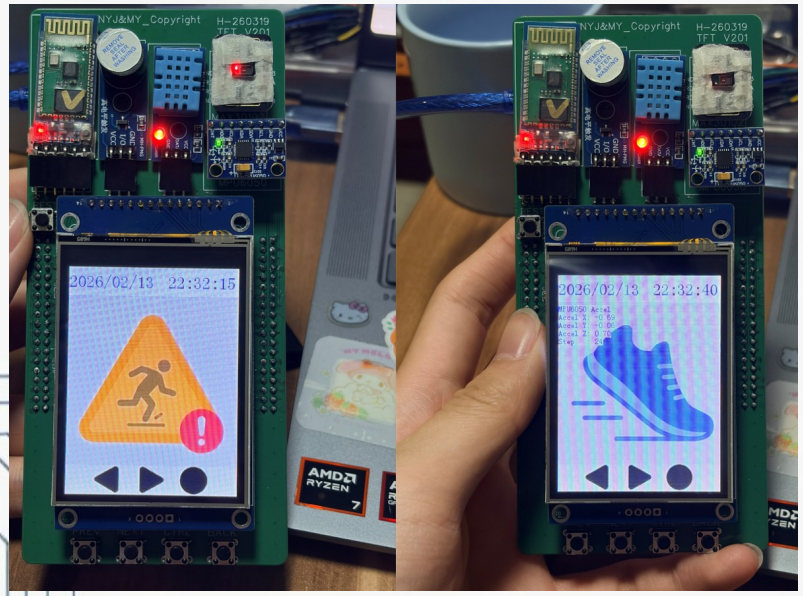
Precautions

- Use a 5V/1A power supply to avoid instability.
- Avoid contact between the sensor and water or other liquids.
- It is not recommended to use the game function for more than 10 minutes at a time.
- It is recommended to contact a technician every six months for sensor calibration.

Safety Tips: Please use the equipment strictly according to the operation manual. If you experience any discomfort, please stop using it immediately and consult a professional physician.

Project Results and Summary





Health monitoring terminal finished product display

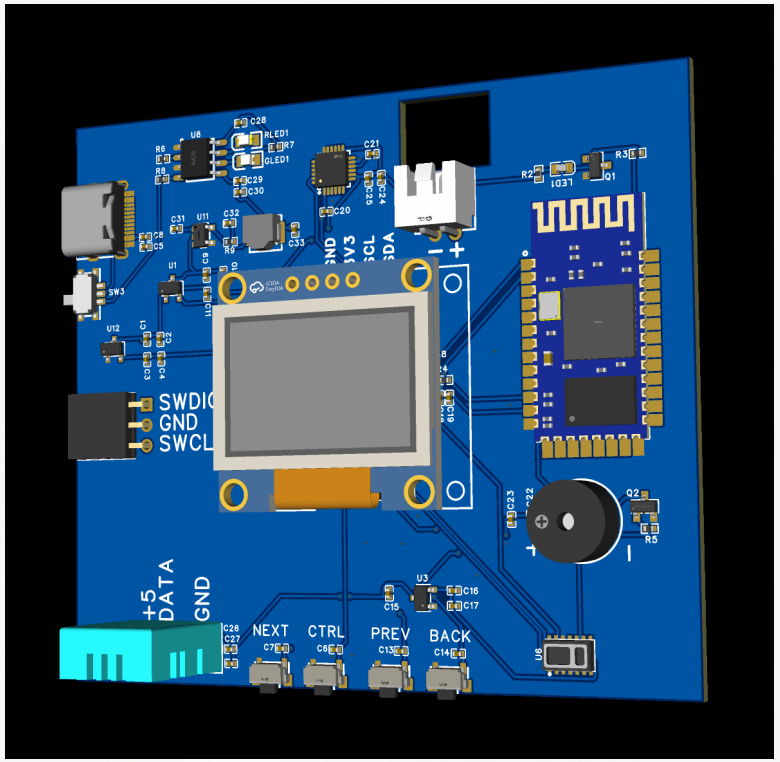
Core product delivery

- It fully realizes five core functions, and its performance indicators fully meet home medical standards.
- Fully realizes five core functions, and its performance indicators fully meet home medical standards.
- Develop supporting PC-based host computer software to enable real-time visual monitoring and historical data recording.
- Completed the main structure design of the equipment, iterated to version 3, and optimized module layout.
- Write project technical reports and user guide, ensuring complete documentation standards.

Course objectives achieved: Project deliverables **100%** meet the learning outcome requirements of the EL2205 course.

One More Thing:

The ultimate integration plan to abandon the development board



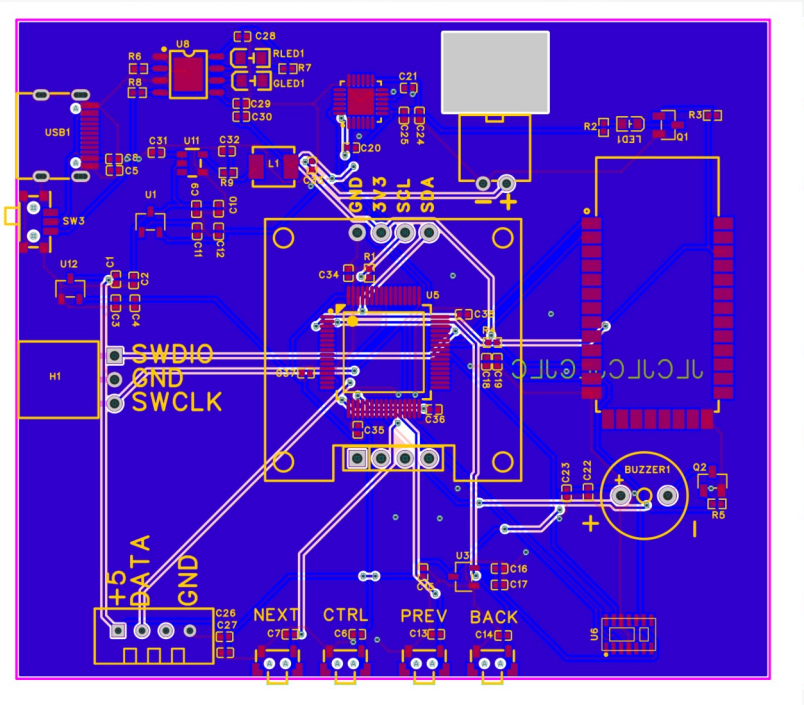
Housing Design: Adaptation and Protection

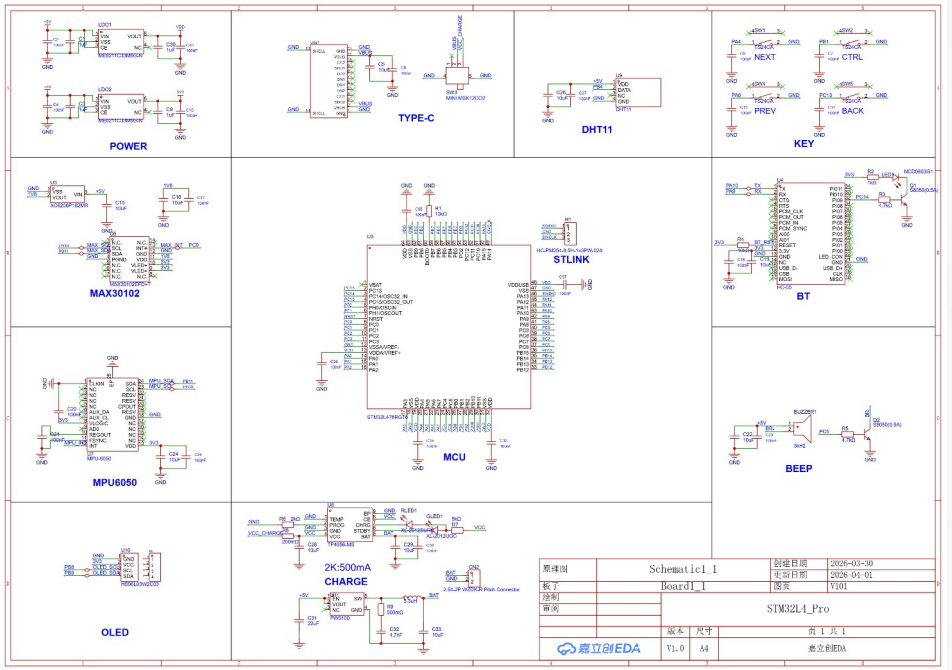
Adopt an upper and lower split structure to perfectly fit the core board and expansion board, and reserve enough space for the battery. Choose PLA material and optimize the edge fillet design to ensure the device's protection and take into account the holding feel at the same time

Iteration Direction: Integrated Wearable

Plan an integrated wristband solution, integrate all functional modules into a single PCB, realize the form upgrade from a handheld prototype to a wrist-worn wearable device, greatly improve portability and user experience, and serve users with outdoor needs.

One More Thing: Schematic & PCB Layout





Schematic diagram

Integrate peripheral circuits of each sensor module, charge and discharge circuit and LDO voltage regulator circuit on one whole PCB board to avoid wasting space.

PCB Layout

Put the core part (stm32) and OLED in the middle, and make the PCB size 80mm*70mm.

Reference Lists

Stroke Association (2024)
Stroke statistics.

NHS England (2019)
The NHS Long Term Plan.

World Health Organization (2021)
Falls.

Johnson, H.W. & Graham, M. (1993)
High-speed digital design: a handbook of black magic.

IPC (2012)
IPC-2221B: Generic standard on printed board design.

STMicroelectronics (2024)
UM1724 User manual: STM32L4 Nucleo-64 boards (MB1136).

Noury, N., et al. (2007)
Fall detection - Principles and Methods. IEEE Engineering in Medicine and Biology Society.

Source: Project development report

Thank you for listening !

Q & A Part

Technical support email:

[redacted]/[redacted]